



7280 - Linking the outer structure with inner chemistry in disks around the most common planet hosts

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		MIRI Medium Resolution Spectroscopy	(1) BCG93-1
	2		MIRI Medium Resolution Spectroscopy	(2) MHO6
	3		MIRI Medium Resolution Spectroscopy	(3) 2MASS-J04334465+2615005
	4		MIRI Medium Resolution Spectroscopy	(4) 2MASS-J04124068+2438157
	5		MIRI Medium Resolution Spectroscopy	(5) 2MASS-J05044139+2509544
	6		MIRI Medium Resolution Spectroscopy	(6) NAME-ZZ-Tau-IRS
	7		MIRI Medium Resolution Spectroscopy	(7) 2MASS-J16000236-4222145
	8		MIRI Medium Resolution Spectroscopy	(8) Sz100
	10		MIRI Medium Resolution Spectroscopy	(10) NAME-Sz-108B
	11		MIRI Medium Resolution Spectroscopy	(11) Sz113
	12		MIRI Medium Resolution Spectroscopy	(12) 2MASS-J16090141-3925119

ABSTRACT

Very low-mass stars (VLMS, spectral type later than M3) are known as favorable hosts of small planets, which are now being intensively studied with JWST. To fully understand the formation, evolution, and chemical compositions of these planets, it is essential to examine the physical and chemical properties of their birthplaces - the disks surrounding VLMS. The exceptional sensitivity of JWST has opened new avenues for investigating these faint VLMS disks, revealing rich and diverse C, O chemistry in their inner regions. One prevailing hypothesis links the variations in gas chemistry to the structure of the outer disk, which controls the inward drift efficiency of icy pebbles and the delivery of water. However, the current JWST VLMS sample is dominated by disks that are faint at millimeter wavelengths, making it difficult to characterize their outer disk structures. To thoroughly test the influence of outer disk structures on inner gas chemistry, we propose MIRI/MRS observations towards a complementary sample of mm-bright disks around VLMS, for which high-resolution ALMA images are already available. By conducting a joint analysis of high-quality JWST and ALMA data, we aim to test if disks with a high inner gas C/O ratio exhibit deep/wide gaps that inhibit pebble drift, while sources with prominent water emission display only shallow features that allow icy pebbles to continue flowing inward.

OBSERVING DESCRIPTION

We request MIRI/MRS observations for 12 disks in the young Taurus and Lupus star-forming regions. We will use all three grating settings to cover the full wavelength range from 5 to 28 micron. As we focus on the detection of faint molecular emission lines, the exposure time for each disk is set to reach SNR of ~ 200 at 17 micron. Each observation will start with target acquisition to place the source at the center of the IFU field of view. The

standard 4-point dither is adopted as the line emission of interest is expected to be unresolved.

Proposal 7280 - Targets - Linking the outer structure with inner chemistry in disks around the most common planet hosts

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	BCG93-1	RA: 04 14 17.6096 (63.5733733d) Dec: +28 06 9.65 (28.10268d) Equinox: J2000	Proper Motion RA: 8.285 mas/yr Proper Motion Dec: -23.60700000281213 mas/yr Parallax: 0.007428300000000005" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Pre-main sequence stars, Protoplanetary disks] Extended=NO				
	(2)	MHO6	RA: 04 32 22.1099 (68.0921246d) Dec: +18 27 42.65 (18.46185d) Equinox: J2000	Proper Motion RA: 12.474 mas/yr Proper Motion Dec: -18.692999992708792 mas/yr Parallax: 0.0068672" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Pre-main sequence stars, Protoplanetary disks] Extended=NO				
Fixed Targets	(3)	2MASS-J04334465+2615005	RA: 04 33 44.6585 (68.4360771d) Dec: +26 15 0.44 (26.25012d) Equinox: J2000	Proper Motion RA: 8.054 mas/yr Proper Motion Dec: -17.74699990164663 mas/yr Parallax: 0.00618" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Pre-main sequence stars, Protoplanetary disks] Extended=NO				
	(4)	2MASS-J04124068+2438157	RA: 04 12 40.6930 (63.1695542d) Dec: +24 38 15.73 (24.63770d) Equinox: J2000	Proper Motion RA: 14.226999999999999 mas/yr Proper Motion Dec: -18.844000101125857 mas/yr Parallax: 0.0067261" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Pre-main sequence stars, Protoplanetary disks] Extended=NO				

Proposal 7280 - Targets - Linking the outer structure with inner chemistry in disks around the most common planet hosts

(5)	2MASS-J05044139+2509544	RA: 05 04 41.3972 (76.1724883d) Dec: +25 09 54.58 (25.16516d) Equinox: J2000	Proper Motion RA: 2.379 mas/yr Proper Motion Dec: -16.979000042738335 mas/yr Parallax: 0.005916999999999995" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Pre-main sequence stars, Protoplanetary disks] Extended=NO			
(6)	NAME-ZZ-Tau-IRS	RA: 04 30 51.7209 (67.7155038d) Dec: +24 41 47.46 (24.69652d) Equinox: J2000	Proper Motion RA: 10.675 mas/yr Proper Motion Dec: -14.991000057307247 mas/yr Parallax: 0.0094606" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Pre-main sequence stars, Protoplanetary disks] Extended=NO			
(7)	2MASS-J16000236-4222145	RA: 16 00 2.3584 (240.0098267d) Dec: -42 22 14.60 (-42.37072d) Equinox: J2000	Proper Motion RA: -11.517 mas/yr Proper Motion Dec: -23.211000097944634 mas/yr Parallax: 0.0062347" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Pre-main sequence stars, Protoplanetary disks] Extended=NO			
(8)	Sz100	RA: 16 08 25.7634 (242.1073475d) Dec: -39 06 1.25 (-39.10035d) Equinox: J2000	Proper Motion RA: -10.26 mas/yr Proper Motion Dec: -22.159000036481302 mas/yr Parallax: 0.0070902000000000005" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Pre-main sequence stars, Protoplanetary disks] Extended=NO			
(10)	NAME-Sz-108B	RA: 16 08 42.8839 (242.1786829d) Dec: -39 06 14.69 (-39.10408d) Equinox: J2000	Proper Motion RA: -10.679 mas/yr Proper Motion Dec: -21.776999915346096 mas/yr Parallax: 0.0062027" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Pre-main sequence stars, Protoplanetary disks] Extended=NO			

Proposal 7280 - Targets - Linking the outer structure with inner chemistry in disks around the most common planet hosts

(11)	Sz113	RA: 16 08 57.8027 (242.2408446d) Dec: -39 02 22.85 (-39.03968d) Equinox: J2000	Proper Motion RA: -9.748 mas/yr Proper Motion Dec: -23.218000092128932 mas/yr Parallax: 0.0062294" Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> <i>Category=Star</i> <i>Description=[Pre-main sequence stars, Protoplanetary disks]</i> <i>Extended=NO</i>			
(12)	2MASS-J16090141-3925119	RA: 16 09 1.4148 (242.2558950d) Dec: -39 25 11.92 (-39.41998d) Equinox: J2000	Proper Motion RA: -9.824 mas/yr Proper Motion Dec: -23.804999955245876 mas/yr Parallax: 0.0062816" Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> <i>Category=Star</i> <i>Description=[Pre-main sequence stars, Protoplanetary disks]</i> <i>Extended=NO</i>			

Proposal 7280 - Observation 1 - Linking the outer structure with inner chemistry in disks around the most common planet hosts

Observation	Proposal 7280, Observation 1													Fri May 23 22:00:09 GMT 2025
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(1)	BCG93-1	RA: 04 14 17.6096 (63.5733733d) Dec: +28 06 9.65 (28.10268d) Equinox: J2000				Proper Motion RA: 8.285 mas/yr Proper Motion Dec: -23.60700000281213 mas/yr Parallax: 0.0074283000000000005" Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
	Category=Star Description=[Pre-main sequence stars, Protoplanetary disks] Extended=NO													
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID				
	1	SAME	FND	FAST	4	1		1	11.1	222662.16				
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction			
	All MRS		YES				FULL				Allow Auto Reorder			
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				POINT SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F770W	FASTR1	30	1	1	Dither 1	4	4	333.005		
	1	SHORT(A)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005		
	1	SHORT(A)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005		
	2		IMAGER	F1000W	FASTR1	30	1	1	Dither 1	4	4	333.005		
	2	MEDIUM(B)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005		
	2	MEDIUM(B)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005		
	3		IMAGER	F1130W	FASTR1	30	1	1	Dither 1	4	4	333.005		
	3	LONG(C)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005		
	3	LONG(C)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005		

Proposal 7280 - Observation 2 - Linking the outer structure with inner chemistry in disks around the most common planet hosts

Observation	Proposal 7280, Observation 2													Fri May 23 22:00:09 GMT 2025	
	Diagnostic Status: Warning														
	Observing Template: MIRI Medium Resolution Spectroscopy														
Diagnostics	(Visit 2:1) Warning (Form): Data Excess over lower threshold														
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.														
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections				Miscellaneous				
	(2)	MHO6		RA: 04 32 22.1099 (68.0921246d)			Proper Motion RA: 12.474 mas/yr								
				Dec: +18 27 42.65 (18.46185d)			Proper Motion Dec: -18.692999992708792 mas/yr								
				Equinox: J2000			Parallax: 0.0068672"								
							Epoch of Position: 2000								
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.														
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.														
	Category=Star														
	Description=[Pre-main sequence stars, Protoplanetary disks]														
	Extended=NO														
Acquisition	#	Target		Filter	Readout Pattern		Groups/Int	Integrations/Exp		Total Integrations		Total Exposure Time		ETC Wkbk.Calc ID	
	1	SAME		FND	FAST		4	1		1		11.1		222662.17	
Template	Primary Channel			Simultaneous Imaging				Imager Subarray				Grating Wheel Direction			
	All MRS			YES				FULL				Allow Auto Reorder			
Dithers	#	Dither Type					Optimized For					Direction			
	1	4-Point					POINT SOURCE					NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID		
	1		IMAGER	F770W	FASTR1	50	1	1	Dither 1	4	4	555.008			
	1	SHORT(A)	MRSLONG		FASTR1	50	4	1	Dither 1	4	16	2253.332			
	1	SHORT(A)	MRSSHORT		FASTR1	50	4	1	Dither 1	4	16	2253.332			
	2		IMAGER	F1000W	FASTR1	50	1	1	Dither 1	4	4	555.008			
	2	MEDIUM(B)	MRSLONG		FASTR1	50	4	1	Dither 1	4	16	2253.332			
	2	MEDIUM(B)	MRSSHORT		FASTR1	50	4	1	Dither 1	4	16	2253.332			
	3		IMAGER	F1130W	FASTR1	50	1	1	Dither 1	4	4	555.008			
	3	LONG(C)	MRSLONG		FASTR1	50	4	1	Dither 1	4	16	2253.332			
	3	LONG(C)	MRSSHORT		FASTR1	50	4	1	Dither 1	4	16	2253.332			

Proposal 7280 - Observation 3 - Linking the outer structure with inner chemistry in disks around the most common planet hosts

Observation	Proposal 7280, Observation 3													Fri May 23 22:00:09 GMT 2025
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(3)	2MASS-J04334465+2615005	RA: 04 33 44.6585 (68.4360771d) Dec: +26 15 0.44 (26.25012d) Equinox: J2000				Proper Motion RA: 8.054 mas/yr Proper Motion Dec: -17.74699990164663 mas/yr Parallax: 0.00618" Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
	Category=Star Description=[Pre-main sequence stars, Protoplanetary disks] Extended=NO													
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID				
	1	SAME	FND	FAST	4	1		1	11.1	222662.17				
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction			
	All MRS		YES				FULL				Allow Auto Reorder			
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				POINT SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F770W	FASTR1	45	1	1	Dither 1	4	4	499.507		
	1	SHORT(A)	MRSLONG		FASTR1	45	1	1	Dither 1	4	4	499.507		
	1	SHORT(A)	MRSSHORT		FASTR1	45	1	1	Dither 1	4	4	499.507		
	2		IMAGER	F1000W	FASTR1	45	1	1	Dither 1	4	4	499.507		
	2	MEDIUM(B)	MRSLONG		FASTR1	45	1	1	Dither 1	4	4	499.507		
	2	MEDIUM(B)	MRSSHORT		FASTR1	45	1	1	Dither 1	4	4	499.507		
	3		IMAGER	F1130W	FASTR1	45	1	1	Dither 1	4	4	499.507		
	3	LONG(C)	MRSLONG		FASTR1	45	1	1	Dither 1	4	4	499.507		
	3	LONG(C)	MRSSHORT		FASTR1	45	1	1	Dither 1	4	4	499.507		

Proposal 7280 - Observation 4 - Linking the outer structure with inner chemistry in disks around the most common planet hosts

Observation	Proposal 7280, Observation 4													Fri May 23 22:00:09 GMT 2025
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(4)	2MASS-J04124068+2438157	RA: 04 12 40.6930 (63.1695542d) Dec: +24 38 15.73 (24.63770d) Equinox: J2000				Proper Motion RA: 14.226999999999999 mas/yr Proper Motion Dec: -18.844000101125857 mas/yr Parallax: 0.0067261" Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
	Category=Star Description=[Pre-main sequence stars, Protoplanetary disks] Extended=NO													
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID				
	1	SAME	FND	FAST	4	1		1	11.1	222662.17				
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction			
	All MRS		YES				FULL				Allow Auto Reorder			
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				POINT SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F770W	FASTR1	50	1	1	Dither 1	4	4	555.008		
	1	SHORT(A)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008		
	1	SHORT(A)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		
	2		IMAGER	F1000W	FASTR1	50	1	1	Dither 1	4	4	555.008		
	2	MEDIUM(B)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008		
	2	MEDIUM(B)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		
	3		IMAGER	F1130W	FASTR1	50	1	1	Dither 1	4	4	555.008		
	3	LONG(C)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008		
	3	LONG(C)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		

Proposal 7280 - Observation 5 - Linking the outer structure with inner chemistry in disks around the most common planet hosts

Observation	Proposal 7280, Observation 5													Fri May 23 22:00:09 GMT 2025
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(5)	2MASS-J05044139+2509544	RA: 05 04 41.3972 (76.1724883d) Dec: +25 09 54.58 (25.16516d) Equinox: J2000				Proper Motion RA: 2.379 mas/yr Proper Motion Dec: -16.979000042738335 mas/yr Parallax: 0.005916999999999995" Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
	Category=Star Description=[Pre-main sequence stars, Protoplanetary disks] Extended=NO													
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID				
	1	SAME	FND	FAST	4	1		1	11.1	222662.17				
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction			
	All MRS		YES				FULL				Allow Auto Reorder			
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				POINT SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F770W	FASTR1	30	1	1	Dither 1	4	4	333.005		
	1	SHORT(A)	MRSLONG		FASTR1	30	2	1	Dither 1	4	8	677.11		
	1	SHORT(A)	MRSSHORT		FASTR1	30	2	1	Dither 1	4	8	677.11		
	2		IMAGER	F1000W	FASTR1	30	1	1	Dither 1	4	4	333.005		
	2	MEDIUM(B)	MRSLONG		FASTR1	30	2	1	Dither 1	4	8	677.11		
	2	MEDIUM(B)	MRSSHORT		FASTR1	30	2	1	Dither 1	4	8	677.11		
	3		IMAGER	F1130W	FASTR1	30	1	1	Dither 1	4	4	333.005		
	3	LONG(C)	MRSLONG		FASTR1	30	2	1	Dither 1	4	8	677.11		
	3	LONG(C)	MRSSHORT		FASTR1	30	2	1	Dither 1	4	8	677.11		

Proposal 7280 - Observation 6 - Linking the outer structure with inner chemistry in disks around the most common planet hosts

Observation	Proposal 7280, Observation 6													Fri May 23 22:00:09 GMT 2025
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(6)	NAME-ZZ-Tau-IRS	RA: 04 30 51.7209 (67.7155038d) Dec: +24 41 47.46 (24.69652d) Equinox: J2000				Proper Motion RA: 10.675 mas/yr Proper Motion Dec: -14.991000057307247 mas/yr Parallax: 0.0094606" Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
	Category=Star Description=[Pre-main sequence stars, Protoplanetary disks] Extended=NO													
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID				
	1	SAME	FND	FAST	4	1		1	11.1	222662.17				
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction			
	All MRS		YES				FULL				Allow Auto Reorder			
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				POINT SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F770W	FASTR1	20	1	1	Dither 1	4	4	222.003		
	1	SHORT(A)	MRSLONG		FASTR1	20	1	1	Dither 1	4	4	222.003		
	1	SHORT(A)	MRSSHORT		FASTR1	20	1	1	Dither 1	4	4	222.003		
	2		IMAGER	F1000W	FASTR1	20	1	1	Dither 1	4	4	222.003		
	2	MEDIUM(B)	MRSLONG		FASTR1	20	1	1	Dither 1	4	4	222.003		
	2	MEDIUM(B)	MRSSHORT		FASTR1	20	1	1	Dither 1	4	4	222.003		
	3		IMAGER	F1130W	FASTR1	20	1	1	Dither 1	4	4	222.003		
	3	LONG(C)	MRSLONG		FASTR1	20	1	1	Dither 1	4	4	222.003		
	3	LONG(C)	MRSSHORT		FASTR1	20	1	1	Dither 1	4	4	222.003		

Proposal 7280 - Observation 7 - Linking the outer structure with inner chemistry in disks around the most common planet hosts

Observation	Proposal 7280, Observation 7													Fri May 23 22:00:09 GMT 2025
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(7)	2MASS-J16000236-4222145	RA: 16 00 2.3584 (240.0098267d) Dec: -42 22 14.60 (-42.37072d) Equinox: J2000				Proper Motion RA: -11.517 mas/yr Proper Motion Dec: -23.211000097944634 mas/yr Parallax: 0.0062347" Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
	Category=Star Description=[Pre-main sequence stars, Protoplanetary disks] Extended=NO													
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID				
	1	SAME	FND	FAST	4	1		1	11.1	222662.17				
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction			
	All MRS		YES				FULL				Allow Auto Reorder			
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				POINT SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F770W	FASTR1	50	1	1	Dither 1	4	4	555.008		
	1	SHORT(A)	MRSLONG		FASTR1	50	2	1	Dither 1	4	8	1121.116		
	1	SHORT(A)	MRSSHORT		FASTR1	50	2	1	Dither 1	4	8	1121.116		
	2		IMAGER	F1000W	FASTR1	50	1	1	Dither 1	4	4	555.008		
	2	MEDIUM(B)	MRSLONG		FASTR1	50	2	1	Dither 1	4	8	1121.116		
	2	MEDIUM(B)	MRSSHORT		FASTR1	50	2	1	Dither 1	4	8	1121.116		
	3		IMAGER	F1130W	FASTR1	50	1	1	Dither 1	4	4	555.008		
	3	LONG(C)	MRSLONG		FASTR1	50	2	1	Dither 1	4	8	1121.116		
	3	LONG(C)	MRSSHORT		FASTR1	50	2	1	Dither 1	4	8	1121.116		

Proposal 7280 - Observation 8 - Linking the outer structure with inner chemistry in disks around the most common planet hosts

Observation	Proposal 7280, Observation 8													Fri May 23 22:00:09 GMT 2025
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(8)	Sz100	RA: 16 08 25.7634 (242.1073475d) Dec: -39 06 1.25 (-39.10035d) Equinox: J2000				Proper Motion RA: -10.26 mas/yr Proper Motion Dec: -22.159000036481302 mas/yr Parallax: 0.0070902000000000005" Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
	Category=Star Description=[Pre-main sequence stars, Protoplanetary disks] Extended=NO													
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID				
	1	SAME	FND	FAST	4	1		1	11.1	222662.17				
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction			
	All MRS		YES				FULL				Allow Auto Reorder			
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				POINT SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F770W	FASTR1	40	1	1	Dither 1	4	4	444.006		
	1	SHORT(A)	MRSLONG		FASTR1	40	1	1	Dither 1	4	4	444.006		
	1	SHORT(A)	MRSSHORT		FASTR1	40	1	1	Dither 1	4	4	444.006		
	2		IMAGER	F1000W	FASTR1	40	1	1	Dither 1	4	4	444.006		
	2	MEDIUM(B)	MRSLONG		FASTR1	40	1	1	Dither 1	4	4	444.006		
	2	MEDIUM(B)	MRSSHORT		FASTR1	40	1	1	Dither 1	4	4	444.006		
	3		IMAGER	F1130W	FASTR1	40	1	1	Dither 1	4	4	444.006		
	3	LONG(C)	MRSLONG		FASTR1	40	1	1	Dither 1	4	4	444.006		
	3	LONG(C)	MRSSHORT		FASTR1	40	1	1	Dither 1	4	4	444.006		

Proposal 7280 - Observation 10 - Linking the outer structure with inner chemistry in disks around the most common planet hosts

Observation	Proposal 7280, Observation 10												Fri May 23 22:00:09 GMT 2025			
	Diagnostic Status: Warning															
	Observing Template: MIRI Medium Resolution Spectroscopy															
Diagnostics	(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.															
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous					
	(10)	NAME-Sz-108B	RA: 16 08 42.8839 (242.1786829d) Dec: -39 06 14.69 (-39.10408d) Equinox: J2000				Proper Motion RA: -10.679 mas/yr Proper Motion Dec: -21.776999915346096 mas/yr Parallax: 0.0062027" Epoch of Position: 2000									
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.															
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.															
	Category=Star Description=[Pre-main sequence stars, Protoplanetary disks] Extended=NO															
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID						
	1	SAME	FND	FAST	4	1		1	11.1	222662.17						
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction					
	All MRS		YES				FULL				Allow Auto Reorder					
Dithers	#	Dither Type				Optimized For				Direction						
	1	4-Point				POINT SOURCE				NEGATIVE						
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID			
	1		IMAGER	F770W	FASTR1	50	1	1	Dither 1	4	4	555.008				
	1	SHORT(A)	MRSLONG		FASTR1	50	2	1	Dither 1	4	8	1121.116				
	1	SHORT(A)	MRSSHORT		FASTR1	50	2	1	Dither 1	4	8	1121.116				
	2		IMAGER	F1000W	FASTR1	50	1	1	Dither 1	4	4	555.008				
	2	MEDIUM(B)	MRSLONG		FASTR1	50	2	1	Dither 1	4	8	1121.116				
	2	MEDIUM(B)	MRSSHORT		FASTR1	50	2	1	Dither 1	4	8	1121.116				
	3		IMAGER	F1130W	FASTR1	50	1	1	Dither 1	4	4	555.008				
	3	LONG(C)	MRSLONG		FASTR1	50	2	1	Dither 1	4	8	1121.116				
	3	LONG(C)	MRSSHORT		FASTR1	50	2	1	Dither 1	4	8	1121.116				

Proposal 7280 - Observation 11 - Linking the outer structure with inner chemistry in disks around the most common planet hosts

Observation	Proposal 7280, Observation 11													Fri May 23 22:00:09 GMT 2025
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(11)	Sz113	RA: 16 08 57.8027 (242.2408446d) Dec: -39 02 22.85 (-39.03968d) Equinox: J2000				Proper Motion RA: -9.748 mas/yr Proper Motion Dec: -23.218000092128932 mas/yr Parallax: 0.0062294" Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
	Category=Star Description=[Pre-main sequence stars, Protoplanetary disks] Extended=NO													
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID				
	1	SAME	FND	FAST	4	1		1	11.1	222662.17				
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction			
	All MRS		YES				FULL				Allow Auto Reorder			
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				POINT SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F770W	FASTR1	40	1	1	Dither 1	4	4	444.006		
	1	SHORT(A)	MRSLONG		FASTR1	40	2	1	Dither 1	4	8	899.113		
	1	SHORT(A)	MRSSHORT		FASTR1	40	2	1	Dither 1	4	8	899.113		
	2		IMAGER	F1000W	FASTR1	40	1	1	Dither 1	4	4	444.006		
	2	MEDIUM(B)	MRSLONG		FASTR1	40	2	1	Dither 1	4	8	899.113		
	2	MEDIUM(B)	MRSSHORT		FASTR1	40	2	1	Dither 1	4	8	899.113		
	3		IMAGER	F1130W	FASTR1	40	1	1	Dither 1	4	4	444.006		
	3	LONG(C)	MRSLONG		FASTR1	40	2	1	Dither 1	4	8	899.113		
	3	LONG(C)	MRSSHORT		FASTR1	40	2	1	Dither 1	4	8	899.113		

Proposal 7280 - Observation 12 - Linking the outer structure with inner chemistry in disks around the most common planet hosts

Observation	Proposal 7280, Observation 12												Fri May 23 22:00:09 GMT 2025	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(12)	2MASS-J16090141-3925119	RA: 16 09 1.4148 (242.2558950d) Dec: -39 25 11.92 (-39.41998d) Equinox: J2000				Proper Motion RA: -9.824 mas/yr Proper Motion Dec: -23.804999955245876 mas/yr Parallax: 0.0062816" Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
	Category=Star Description=[Pre-main sequence stars, Protoplanetary disks] Extended=NO													
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID				
	1	SAME	FND	FAST	4	1		1	11.1	222662.17				
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction			
	All MRS		YES				FULL				Allow Auto Reorder			
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				POINT SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F770W	FASTR1	50	1	1	Dither 1	4	4	555.008		
	1	SHORT(A)	MRSLONG		FASTR1	50	2	1	Dither 1	4	8	1121.116		
	1	SHORT(A)	MRSSHORT		FASTR1	50	2	1	Dither 1	4	8	1121.116		
	2		IMAGER	F1000W	FASTR1	50	1	1	Dither 1	4	4	555.008		
	2	MEDIUM(B)	MRSLONG		FASTR1	50	2	1	Dither 1	4	8	1121.116		
	2	MEDIUM(B)	MRSSHORT		FASTR1	50	2	1	Dither 1	4	8	1121.116		
	3		IMAGER	F1130W	FASTR1	50	1	1	Dither 1	4	4	555.008		
	3	LONG(C)	MRSLONG		FASTR1	50	2	1	Dither 1	4	8	1121.116		
	3	LONG(C)	MRSSHORT		FASTR1	50	2	1	Dither 1	4	8	1121.116		